PROJECT 36 | DBMS REVIEW 1

EMPLOYEE PAYROLL AND STATUTORY DEDUCTION MANAGEMENT SYSTEM

A Relational Database Management System for Automated Payroll Processing



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Introduction & Problem Identification

Organizations must accurately manage employee, salary, attendance, earnings, deductions, loan and payroll records throughout every payroll cycle. When this data is maintained on spreadsheets rather than a structured database, the process becomes error-prone, difficult to audit, and hard to scale as the workforce grows.

Manual / Spreadsheet Process

Errors, duplication, and difficult tracking across scattered files.

Centralized Database

Organized, accurate, and easily accessible payroll records.

KEY PROBLEMS IDENTIFIED

Manual Calculation Errors

Salary and deduction totals computed by hand are prone to mistakes.

Scattered Employee Records

Data spread across multiple files with no central source of truth.

Difficult Deduction / Loan Tracking

Hard to monitor statutory deductions and loan repayments accurately.

Payroll-History Management

Retrieving and maintaining past payroll cycles becomes cumbersome.

Difficult Reporting

Generating consolidated payroll reports is slow and unreliable.

Objectives

1

Centralized Payroll Database

Single, unified database for all payroll-related information.

2

Employee, Department & Designation Management

Maintain organizational structure and staff records.

3

Salary Structure Management

Define and manage pay components for every employee.

4

Attendance, Earnings & Deductions

Track attendance and compute earnings and deductions.

5

Loan & Repayment Tracking

Record employee loans and monitor repayment schedules.

6

Payroll Period & Payslip Management

Generate payroll runs and detailed employee payslips.

7

Bank Transfers & Data Integrity

Maintain transfer records while ensuring consistent, accurate data.

Scope and Users

PROJECT SCOPE

- Employee management
- Department management
- Designation management
- Salary structures
- Attendance summaries
- Earnings records
- Deduction records
- Loan management
- Payroll processing
- Payslip generation
- Bank transfer records
- Payroll reports

SYSTEM USERS

HR / Administrator

Manages employee, department and designation records.

Payroll Officer

Processes salary, deductions, loans and payroll runs.

Employee

Views personal attendance, earnings and payslips.

Management

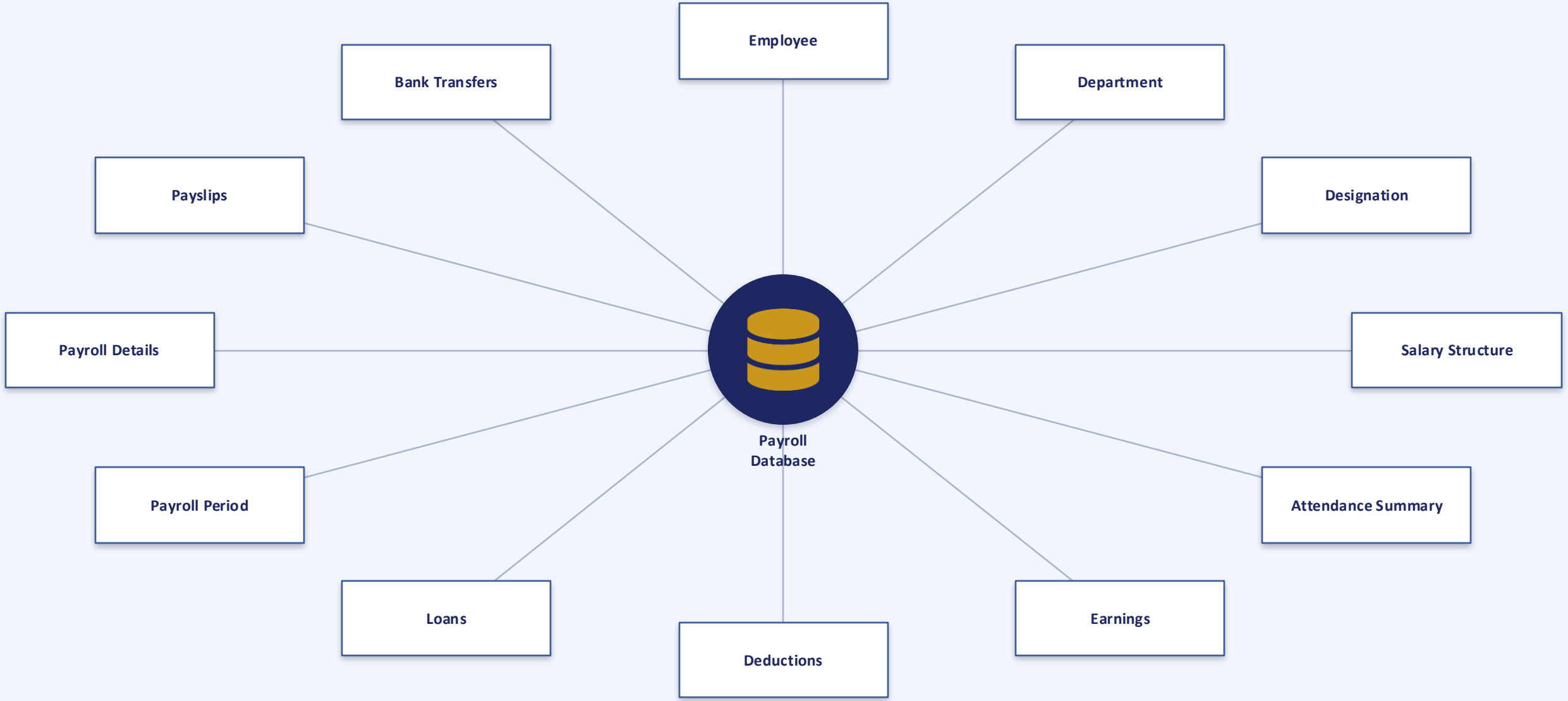
Reviews payroll reports and organization-wide summaries.

Functional Requirements

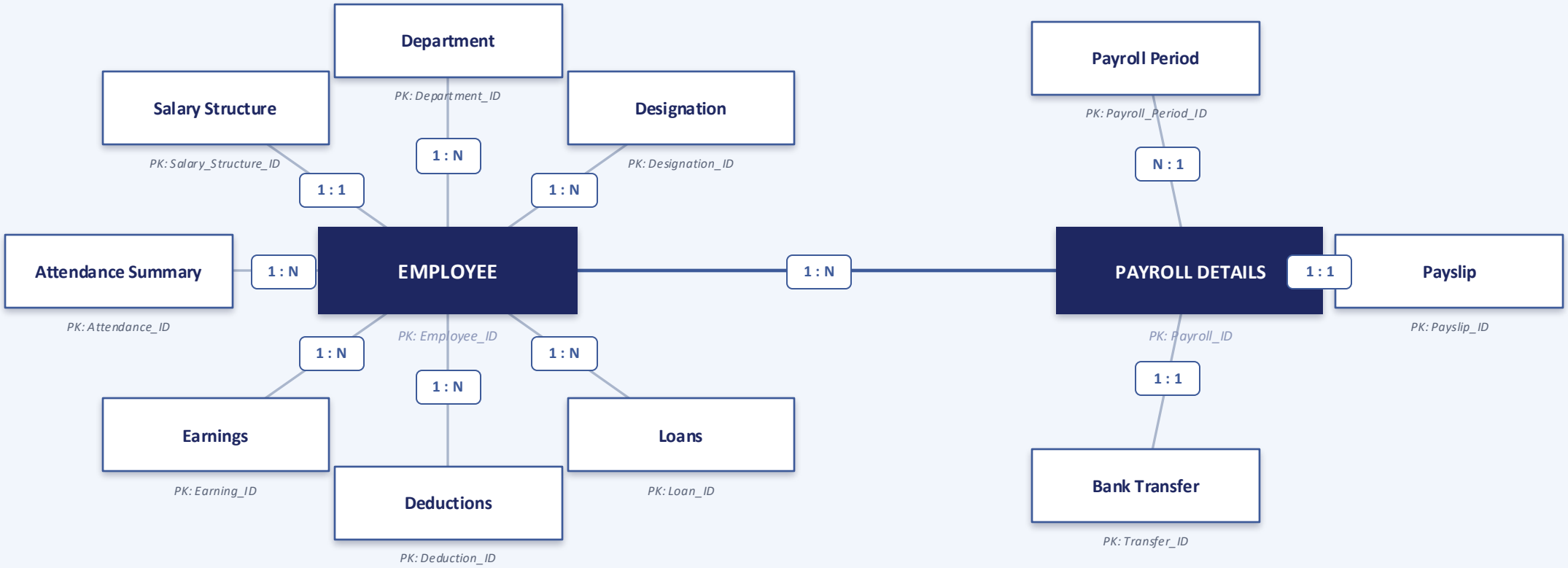


1. Add, update, delete and view employee records
2. Manage department and designation records
3. Maintain employee salary structures
4. Record monthly attendance summaries
5. Record employee earnings
6. Record statutory and other deductions
7. Maintain employee loans and repayments
8. Create payroll records for each payroll period
9. Calculate gross pay, deductions and net pay
10. Generate employee payslips
11. Record bank transfer details
12. Generate payroll-related reports

Principal Entities



Conceptual ER Diagram



Legend: 1 — one | N — many | PK — primary key

Initial Relational Schema & Review Summary

Department
- PK Department_ID

Designation
- PK Designation_ID

Employee
- PK Employee_ID
- FK Department_ID
- FK Designation_ID

Salary Structure
- PK Salary_Structure_ID

Attendance Summary
- PK Attendance_ID
- FK Employee_ID

Earnings
- PK Earning_ID
- FK Employee_ID

Deductions
- PK Deduction_ID
- FK Employee_ID

Loan
- PK Loan_ID
- FK Employee_ID

Payroll Period
- PK Payroll_Period_ID

Payroll Detail
- PK Payroll_ID
- FK Employee_ID
- FK Payroll_Period_ID

Payslip
- PK Payslip_ID
- FK Payroll_ID

Bank Transfer
- PK Transfer_ID
- FK Payroll_ID

REVIEW 1 — COMPLETED

- Problem Identification
- Scope
- Objectives
- Users
- Functional Requirements
- ER Design
- Relational Schema

Thank You